Crime and What Else? Investigating the Predictive Power of Social Factors on Incarceration Rates

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Abstract

Crime is often taken for granted as the primary driver of incarceration in the United States, without accounting for underlying factors (e.g. poverty) or large-scale policies that expand the prison-industrial complex. This project aims to evaluate the truth of this dominant narrative by utilizing machine learning to assess how well a variety of social factors are able to predict incarceration rates. Among both single-feature models and a model using all features to generate predictions, police spending had the lowest error rate and highest feature importance, indicating that government spending on police is more closely associated with incarceration rates than crime.

Introduction & Motivation

Problem & question : As of 2020, the United States prison system held over one million non-federal prisoners, an increase of over 200% in the raw prison population since 1980 [1]. Rather than simply accepting the US's continued reliance on incarceration and its associated harms, we could all benefit from considering how the criminal legal system actually works and whose interests it protects. To that end, **this project seeks to interrogate whether incarceration is better predicted by crime or by social factors (such as race, poverty, or government spending on police)**.

Impact & stakeholders : The key stakeholders in this work are those who are most often targeted by criminalization and incarceration, primarily poor and BIPOC folks. Additionally, communities across the country are continually harmed by increasing incarceration rates, as incarceration has been shown to negatively impact both community health and poverty [2]. By critically examining the circumstances that can contribute to incarceration, this project seeks to uplift the work of organizations such as Transformative Justice Community and the Oregon Justice Resource Center, which advocate for solutions to harm that shift our paradigm away from imprisonment.

Background & Related Work

Data science and statistics have played an important role in prison reform and abolition causes; as just one example, researchers have known for well over three decades that the population of Black prisoners is disproportionate to the Black population in the US [2]. Data concerning overall prison populations have also been key in tracking the effects of legislation concerning crime and sentencing, many of which contributed to astronomical growth of the US prison population [3]. Police spending is another facet of prior research, particularly when it comes to questions of how to effectively and durably address crime—while increased policing may reduce crime in the short term, studies suggest that diverting money from policing into social services more reliably reduces crime in the long term [4].

To this point, most of the work applying machine learning algorithms to data concerning the prison system has focused on predicting or classifying individual behaviors or risk factors. This project takes a more abstracted view of the legal system and its contributing factors as a whole.

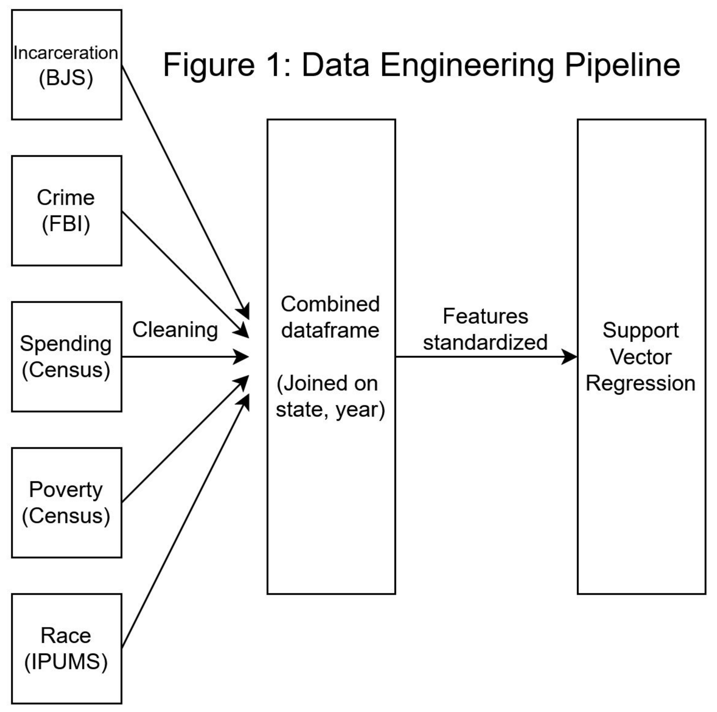
Data, Engineering & Ethics

Sources & scale :
- The data set for this project was constructed by combining data from 5 sources:
  - The Bureau of Justice's Corrections Statistical Analysis Tool
  - The FBI's Summary Reporting System
  - Government spending data from the US Census via the Urban Institute
  - Historical poverty statistics from the Census
  - Race data from IPUMS' Current Population Survey
- **Data span 40 years, from 1980-2020; all US states are represented each year**
  - Total of 2,050 rows

Pipeline :
- All data were ingested as a one-time manual download or loaded directly into R
- Cleaning standardized the format of each dataset
- **Sources were combined into one frame based on state-year pairings**
- Rates per 100k people were calculated based on Census population data

Ethics & responsible use :
- Most of the data used for this project were already highly aggregated or were manually aggregated in cleaning
- Combined data pose little (if any) privacy risks to individuals
  - Consent was a pre-requisite to participate in original surveys
- This is a correlational study, from which no causal claims should be made

Figure 1: Data Engineering Pipeline



Methods & Evaluation

Approach:
- **Support vector regression (SVR)** was used as the primary method
  - Chosen for ability to deal with complex but relatively small data
  - All models were created using default parameters to ensure all variables were treated the same by the algorithm
- **Five single-feature models** trained on individual predictor variables across time, using the national incarceration rate as the label for each
- **A final model was trained on all features**, which was used to find the feature importance of each variable.

Validation :
- Prior to training:
  - **Correlation coefficient** was used as a baseline to assess linear relationships between variables
  - All features were standardized using a robust scaler, split into training/testing sets (80% training, 20% testing)
- After predictions made:
  - **R^2 scores** were used to evaluate how much variance is explained by each feature
  - **Root mean squared error (RMSE)** was found for each, denoting average error in predictions

Results

- **All variables had weak (less than 0.5) correlation** with incarceration rates

Single-feature models:
- All performed poorly; only two features (**police spending at 69.83 and property crime rate at 79.95**) had RMSEs of less than 100 people per 100k
- **R2 scores, none of which were above 0.6**, indicate that no individual feature was able to explain the variance in the national incarceration rate particularly well
- Out of these poor models, **police spending had the lowest RMSE and highest R2 score**

Multi-feature model:
- Performed markedly better: **RMSE of 53.65, R2 of 0.74**
- **Police spending was the most important feature** in generating the model's predictions by two different measures—it had a **permutation importance of 0.24 and SHAP value of 0.21**

Figure 2: National Incarceration Rate Predictions Using All Features and Police Spending Only

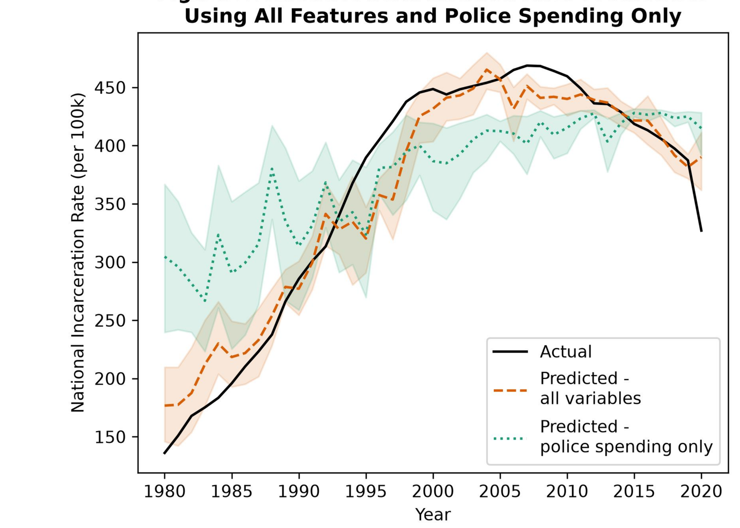


Figure 3: Features are Worse Predictors on Their Own Than When Combined

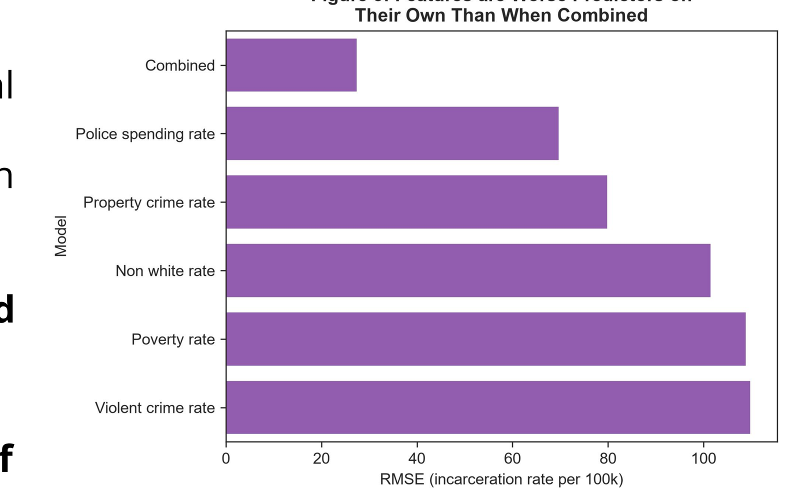


Table 1: All Model Results

Feature	Correlation	RMSEs	R2 scores	Feature importance	SHAP values
Violent crime rate	0.31	109.92	-0.08	0.15	0.11
Property crime rate	-0.08	79.95	0.43	0.14	0.12
Police spending rate	0.42	69.83	0.57	0.24	0.21
Poverty rate	0.15	108.95	-0.06	0.06	0.06
Non white rate	0.46	101.65	0.08	0.11	0.08

Discussion, Limitations & Conclusions

Discussion : The results above indicate that across the board, **police spending rate serves as the best (or most important) for national incarceration rate** out of the variables tested using SVR models. In all evaluation metrics, **property crime rates were a close second**, meaning that where crime was associated with incarceration rates in this project it was primarily property crime and *not* violent crime. Additionally, the performance difference between single-feature and combined-feature models confirms that **none of these variables truly exist in isolation**, and all must be accounted for in order to build a fuller picture of what might influence incarceration in the United States.

Limitations & future work : The chief limitation of this project is that these data and the features used are by nature interconnected; **some of the features used to generate predictions may have impacted others** (such as police spending and crime rates). Additionally, the data used were not objective counts; most are estimates derived from surveys or reports. Future work could explore variations in the type of incarceration (e.g. jails vs. prisons, men vs. women) or concentrate on smaller geographical areas such as counties to incorporate more data.

Conclusions : Police spending served as the most influential predictor of national incarceration rates. Any efforts to reduce incarceration should keep police and legal system spending in mind as a factor when engineering reforms.

Reproducibility & References

Repository: [https://github.com/eskerani/prison-predictor-capstone] · Key dependencies: [Python 3.11, pandas, scikit-learn, seaborn, numpy, matplotlib, dataframe_image; R 4.5.2, tidyverse, readxl, ipumsr]

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5. Sarah Flood, Miriam King, Renae Rodgers, Steven Ruggles, J. Robert Warren, Daniel Backman, Etienne Breton, Grace Cooper, Julia A. Rivera Drew, Stephanie Richards, David Van Riper, and Kari C.W. Williams. IPUMS CPS: Version 13.0 [dataset]. Minneapolis, MN: IPUMS, 2025. https://doi.org/10.18128/D030.V13.0
6. For further information on the other data sources, see the QR code linked to the right.